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QUESTION 1: Beynon-Davies is quoted in the September 1999 issue of *The Computer Bulletin* as saying that “information systems is currently at best a semi-profession”.

Discuss this statement by comparing the information systems development profession with other professions such as medicine and law. [10]

To answer this question, you need to say what the characteristics of a profession are and then discuss how far information systems development shows these characteristics.

The introduction to this chapter lists four characteristics:

1. Substantial education and training are required in order to practice the profession;
2. The members of the profession themselves decide the nature of this training and, more generally, control entry to the profession the profession is organized into one or more professional bodies;
3. The profession lays down standards of conduct with which its members must comply, and where necessary, enforces these through disciplinary procedures

All of these characteristics apply to law and medicine. In both cases, to be registered as a member of the profession, you must pass a lengthy sequence of exams, either set by the professional bodies or set by institutions accredited by them. The professional bodies (the Law Society, the Bar Council, the General Medical Council, and the Royal Colleges) also lay down the level of practical experience required for registration and set and enforce codes of conduct. None of these characteristics apply to information systems development in general. There are very large numbers of people who develop information systems who have no formal education or training in the field; they do not belong to any professional body and are subject to no code of conduct. It cannot therefore be said that information systems development is a profession. However, those information systems developers who belong to professional bodies such as the BCS or the IEE (Institution of Electrical Engineers) can be regarded as constituting a profession, in that membership of those bodies does demonstrate the characteristics identified above, although the ‘experience only’ route would be regarded by many as jeopardizing the claim to professional status.

QUESTION 2: What are Management Buyouts? What are potential conflict of interest(s) in practice of a Management Buyout? [5]

It may happen that some of the senior employees of a company decide to purchase the company from its existing owners. This is known as a management buyout. It will usually require a lot of capital to do this and the purchasers will need to borrow this. However, the lenders are likely to demand that the purchasers shoulder a significant part of the risk by raising a substantial amount of capital, for example, by mortgaging their homes. At first sight a management buyout looks praiseworthy – the people who are running the company become its owners. In practice, there are many potential conflicts of interest. The management may run the company down over a period before bidding to buy it, thus reducing its apparent value and the sum they have to pay to buy it. Once they have bought it, they rapidly build the company up again before selling it at a considerable profit.

QUESTION 3: Explain Centralization/Decentralization and Depth of a Structure with respect to professionals of IT working in an organization. [5]

Organizations may be centralized or decentralized. In a centralized company, as much power as possible is kept at the top of the company, with delegation only when essential. In a decentralized company, as much power and control as possible is delegated to the lowest level. If we take a software company as an example, centralization might mean that there were company-wide rules that all programming should be done in C++ and that, whenever a database package was needed, Oracle should be used. Such a policy has the obvious advantages that programmers could be easily moved from one part of the company to another and that it would be possible to build up a close relationship with Oracle. On the other hand, it might mean that C++ and Oracle were used for projects that would have been much better done using Java and MySQL, or Visual Basic and Access.

Decentralization would allow the most suitable tools to be chosen for each project but might mean that the staff were very inflexible. It could also lead to a maintenance nightmare in the future, with maintenance staff needing to be familiar with large numbers of obsolete tools.

Drawing the correct balance between centralization and decentralization is important but difficult. Decentralization is commonly found in hi-tech companies, where there is plenty of talent at lower levels. Centralization is commoner in large manufacturing companies and other long-established organizations.

The ideal might be described as **flexible centralization**, in which rules and practices are laid down centrally but it is accepted that reasonable arguments for modifying them in specific cases will be readily accepted. Unfortunately, putting this into practice often proves difficult.